SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI – 602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – EXPLORATORY DATA ANALYSIS

Course Code:	DSA05	**Course Title:**	Query Processing for Data Science in Real Time Applications
CO Assessed:	CO4	**Assessment:**	Tool 1 – Exploratory Data Analysis
Weightage:		**Total Marks:**	25 Marks
CO4 Statement: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.. (BL4 and BL5)			
Student Name:	**P.Sunayana**	**Reg. No.:**	**192424303**
Section / Batch:	**Slot-A 2024**	**Date of Submission:**	**19-08-2026**
Faculty In-charge:	K. Veena Devi	**Assessment Mode:**	Individual Submission

Code of Conduct

I _**P.Sunayana (192424303)**_ (Name / Reg No) certify that this submission is my original work and that I have adhered to all guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature: _P.Sunayana_

Dataset Exploration (05 Marks)	Data Cleaning and Preparation (05 Marks)	Statistical Analysis (05 Marks)	Data Visualization (05 Marks)	Insights, Interpretation and Reporting (05 Marks)	Total (25 Marks)

1. Titanic Passenger Survival Analysis

Dataset: Titanic Dataset

Problem Statement:
Explore passenger characteristics and determine factors associated with survival.

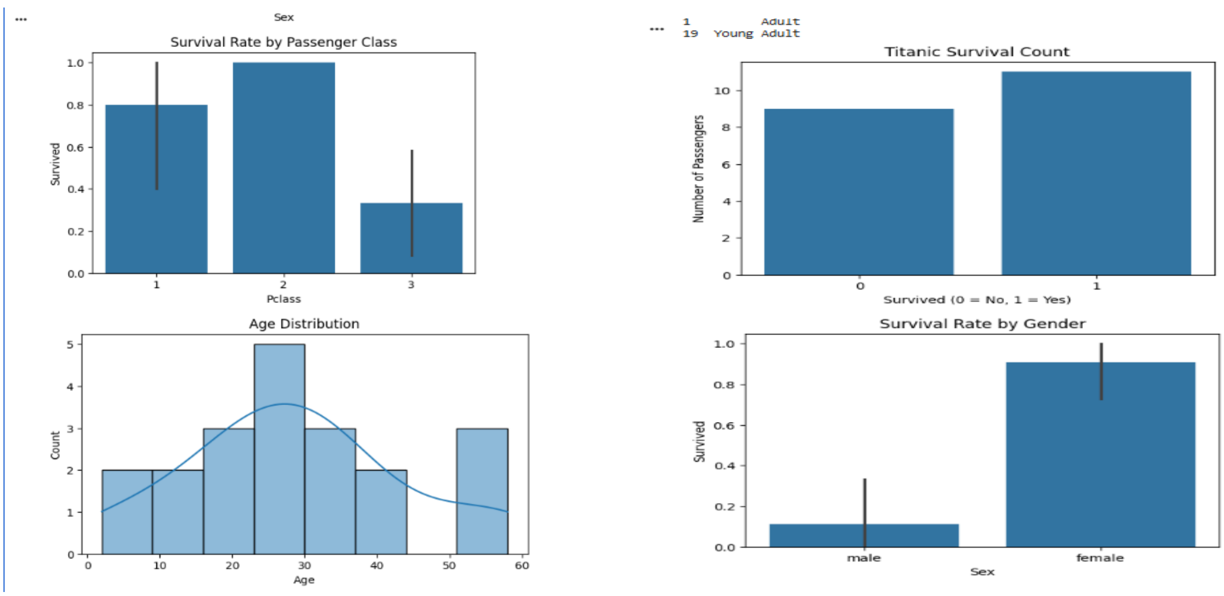
Tasks:

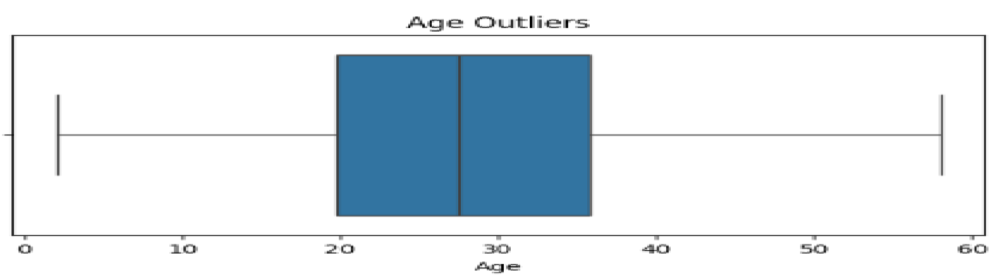
- Handle missing values.
- Analyze survival rates by gender, age, and passenger class.
- Detect outliers in age and fare.
- Visualize categorical and numerical variables.
- Summarize key findings.

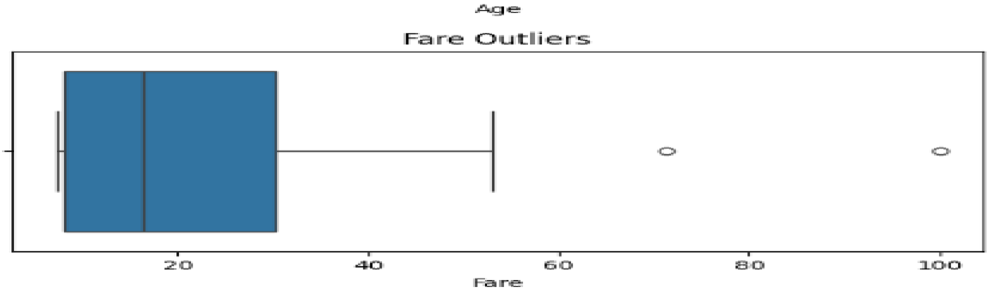
Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.DataFrame({

    "Sex": ["Male","Female","Female","Male","Female","Male"],

    "Age": [22,38,26,35,28,40],

    "Class": [3,1,3,1,2,3],

    "Survived": [0,1,1,1,1,0],

    "Fare": [7,71,8,53,30,10]})

print(df.groupby("Sex")["Survived"].mean())

print(df.groupby("Class")["Survived"].mean())

sns.barplot(data=df, x="Sex", y="Survived")

plt.title("Survival by Gender")
```

```
plt.show()

sns.boxplot(x=df["Fare"])

plt.title("Fare Outliers")

plt.show()
```








```
KEY FINDINGS:
1. Survival differs between male and female passengers.
2. Passenger class is associated with survival rate.
3. Age and fare distributions can contain outliers.
```

Objective: Explore passenger characteristics and identify factors associated with survival.

Main analysis:

- Handle missing values in **Age, Cabin, and Embarked**.
- Calculate survival rates by **Gender**.
- Analyze survival across **Passenger Class (Pclass)**.
- Compare survival rates among different **Age Groups**.
- Detect outliers in **Age** and **Fare** using box plots or the IQR method.
- Visualize categorical variables using:
  - Bar charts
  - Count plots
  - Pie charts
- Visualize numerical variables using:
  - Histograms
  - Box plots
  - Density plots
- Study relationships such as:
  - Gender vs Survival
  - Class vs Survival
  - Age vs Survival

Expected finding: Gender, passenger class, age, and fare are likely to be important factors associated with survival.

2. Employee Attrition Analysis

Dataset: IBM HR Employee Attrition Dataset

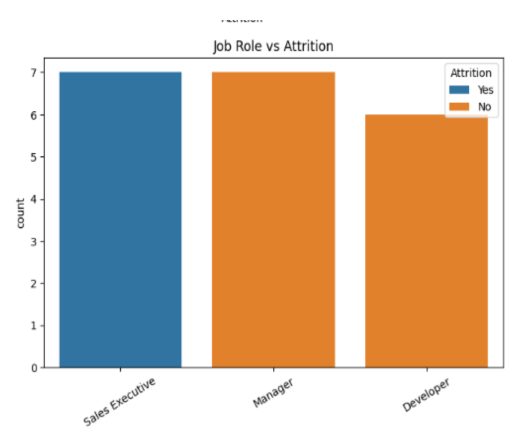
Problem Statement:
Investigate employee attrition patterns and identify major contributing factors.

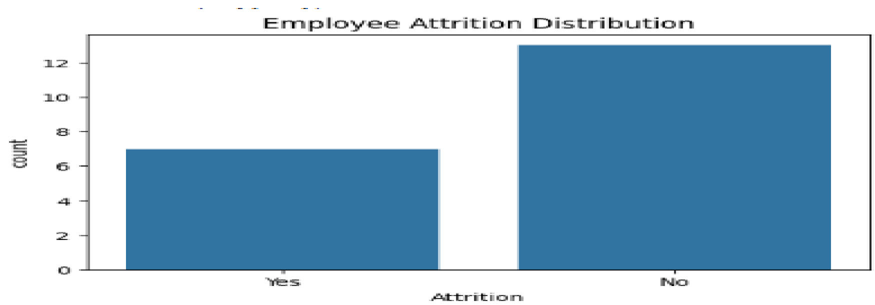
Tasks:

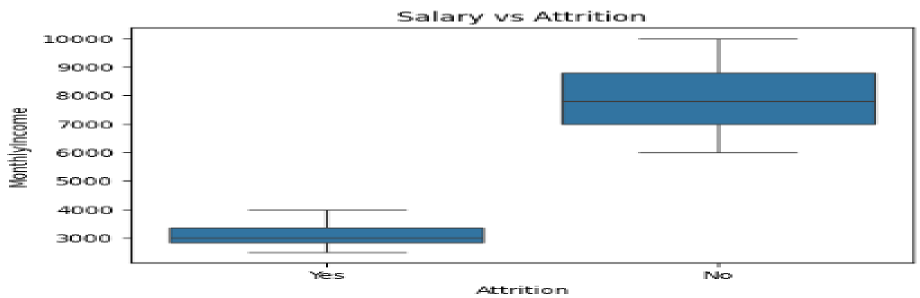
- Examine attrition distribution.
- Compare salary, job role, and years at company.
- Analyze relationships among employee features.
- Identify trends using visualizations.
- Provide recommendations to reduce attrition.

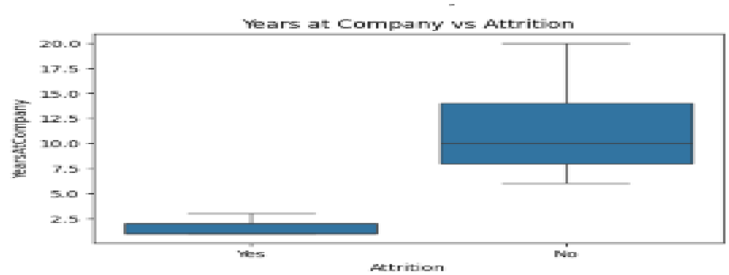
Code:

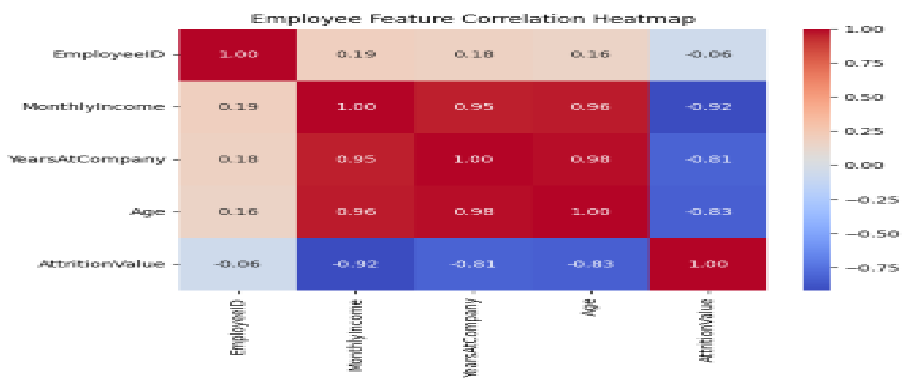
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.DataFrame({

    "Attrition": ["Yes","No","No","Yes","No","Yes"],

    "Salary": [3000,7000,8000,2500,9000,3500],

    "Years": [1,8,10,2,12,3],

    "JobRole": ["Sales","Manager","Developer","Sales","Manager","Developer"]})

print(df["Attrition"].value_counts())

print(df.groupby("Attrition")["Salary"].mean())

sns.countplot(data=df, x="Attrition")

plt.title("Employee Attrition")

plt.show()

sns.boxplot(data=df, x="Attrition", y="Salary")

plt.show()
```











Objective: Investigate employee attrition patterns and identify factors contributing to employees leaving the organization.

Main analysis:

- Examine the distribution of Attrition (Yes/No).
- Compare Monthly Income between employees who stayed and left.
- Analyze attrition by Job Role.
- Compare Years at Company with attrition.

- Analyze relationships among numerical employee features using a correlation heatmap.
- Create visualizations such as:
    - Attrition count plot
    - Salary vs Attrition box plot
    - Job Role vs Attrition bar chart
    - Years at Company distribution
    - Correlation heatmap

Recommendations:

- Improve employee retention strategies.
- Review salary and compensation structures.
- Provide career growth opportunities.
- Focus on job roles with high attrition.
- Support employees during their early years in the company.

3. Housing Price Analysis

Dataset: House Prices Dataset

Problem Statement:
Explore housing market trends and identify factors affecting property prices.

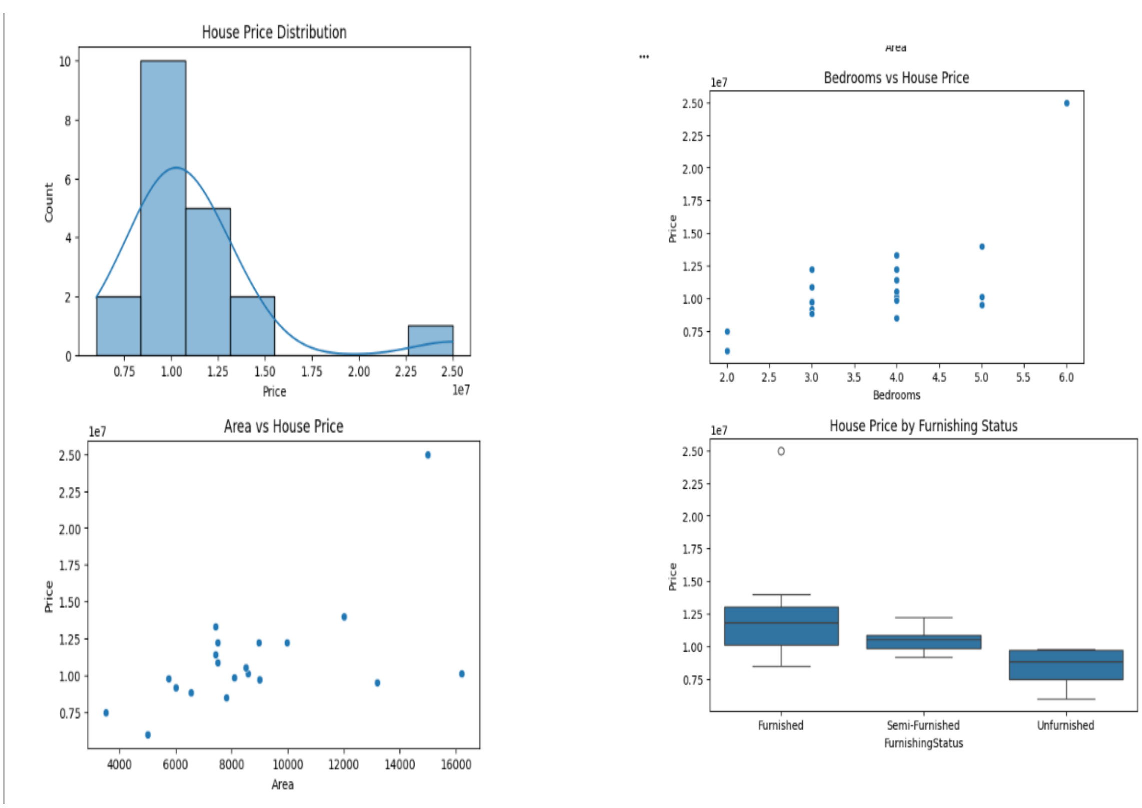
Tasks:

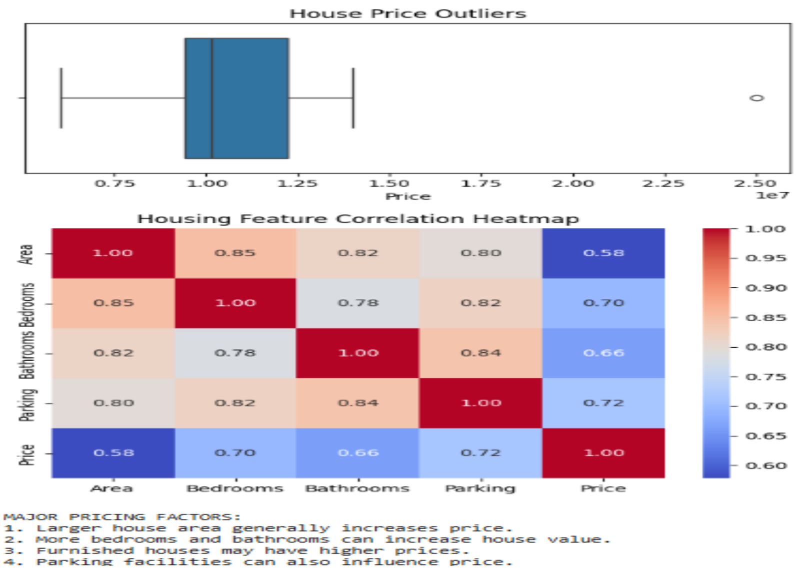
- Analyze numerical and categorical features.
- Detect outliers in house prices.
- Study relationships between area, bedrooms, and price.
- Create correlation heatmaps.
- Summarize major pricing factors.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.DataFrame({

    "Area": [1000,1500,2000,2500,3000,5000],
```

```
    "Bedrooms": [2,3,3,4,4,5],

    "Price": [2000000,3500000,4500000,6000000,7000000,15000000]})

print(df.describe())

sns.scatterplot(data=df, x="Area", y="Price")

plt.title("Area vs Price")

plt.show()

sns.boxplot(x=df["Price"])

plt.title("Price Outliers")

plt.show()

sns.heatmap(df.corr(), annot=True)

plt.show()
```





Objective: Identify the major factors influencing house prices.
Main analysis:
Analyze numerical features such as:

- House Area
- Bedrooms
- Bathrooms
- Parking
- Price

- Analyze categorical features such as:
  - Furnishing Status
  - Main Road
  - Guest Room
  - Basement
- Detect outliers in Price using box plots and IQR.
- Study relationships between:
  - Area vs Price
  - Bedrooms vs Price
  - Bathrooms vs Price
- Create a correlation heatmap.
- Use:
  - Histograms
  - Scatter plots
  - Box plots
  - Bar charts

Expected finding: House area, number of bedrooms/bathrooms, parking, and furnishing status can significantly influence property prices.

4. COVID-19 Data Exploration

Dataset: COVID-19 Global Cases Dataset

Problem Statement:
Analyze COVID-19 statistics across countries and identify significant trends.

Tasks:

- Calculate summary statistics.
- Compare cases, deaths, and recoveries.
- Identify countries with highest infection rates.
- Visualize trends over time.
- Generate actionable insights.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.DataFrame({

    "Country": ["India","USA","Brazil","India","USA","Brazil"],

    "Cases": [1000,2000,1500,2000,4000,3000],

    "Deaths": [20,40,30,35,80,60],

    "Recovered": [700,1300,900,1500,3000,2000]})

print(df.describe())

sns.barplot(data=df, x="Country", y="Cases")

plt.title("COVID Cases by Country")
```

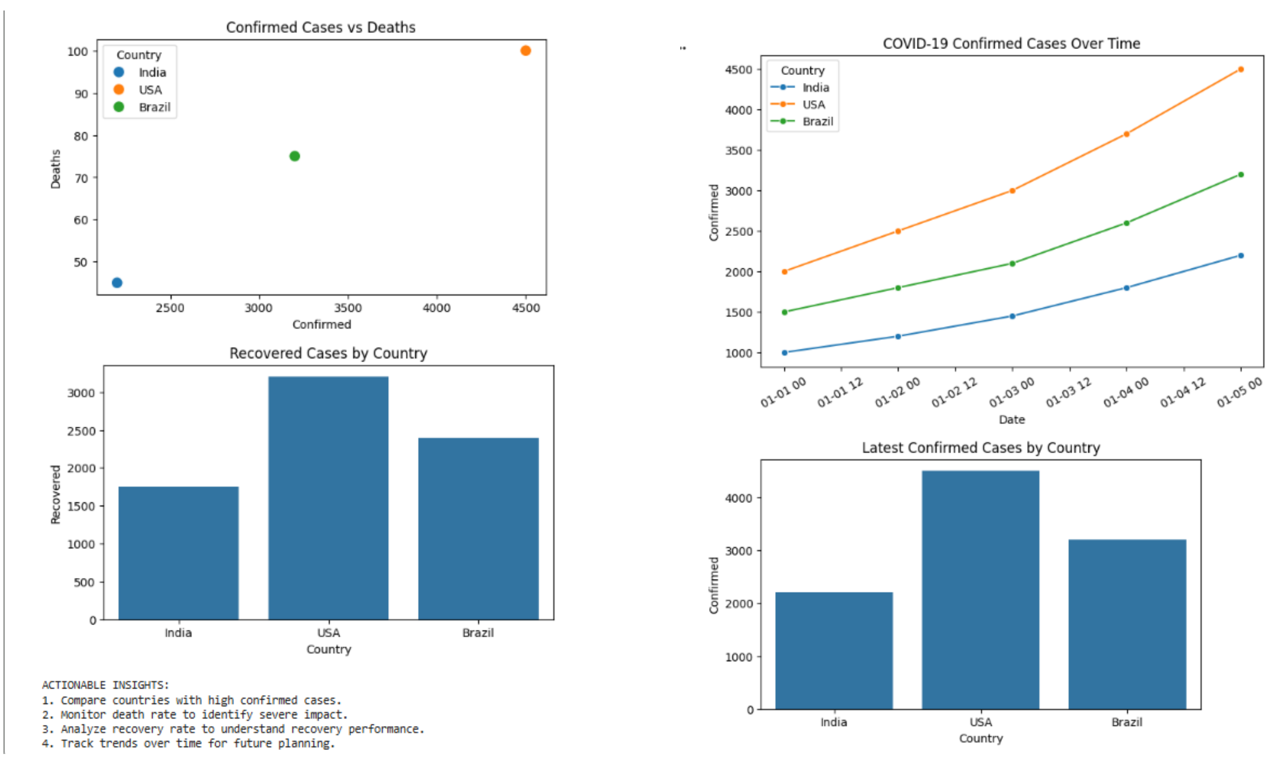
plt.show()

sns.scatterplot(data=df, x="Cases", y="Deaths")

plt.show()

Objective: Analyze global COVID-19 statistics and identify major trends across countries and time.
Main analysis:

- Calculate summary statistics for:
    - Confirmed Cases
    - Deaths
    - Recovered Cases
- Compare cases, deaths, and recoveries across countries.
- Identify countries with the highest:
    - Total cases
    - Death counts
    - Recovery counts
- Calculate infection and death rates where population data is available.
- Visualize trends over time using line charts.
- Create:



    - Bar charts for top affected countries
    - Line charts for time trends

- Scatter plots comparing cases and deaths

Actionable insights:

- Identify countries with high infection growth.
- Compare recovery performance.
- Identify regions with relatively high death rates.
- Support data-driven public health planning.

5. Automobile Market Analysis

Dataset: Automobile MPG Dataset

Problem Statement:
Investigate vehicle characteristics and fuel efficiency trends.

Tasks:

- Explore fuel consumption distribution.
- Compare mileage across vehicle types.
- Analyze relationships among engine size, weight, and MPG.
- Detect anomalies.
- Summarize findings with visualizations.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

df = pd.DataFrame({

    "MPG": [18,25,30,15,35,20],

    "Weight": [3500,2500,2000,4000,1800,3000],

    "Engine": [350,200,150,400,120,250],

    "Cylinders": [8,6,4,8,4,6]})

print(df.describe())
```

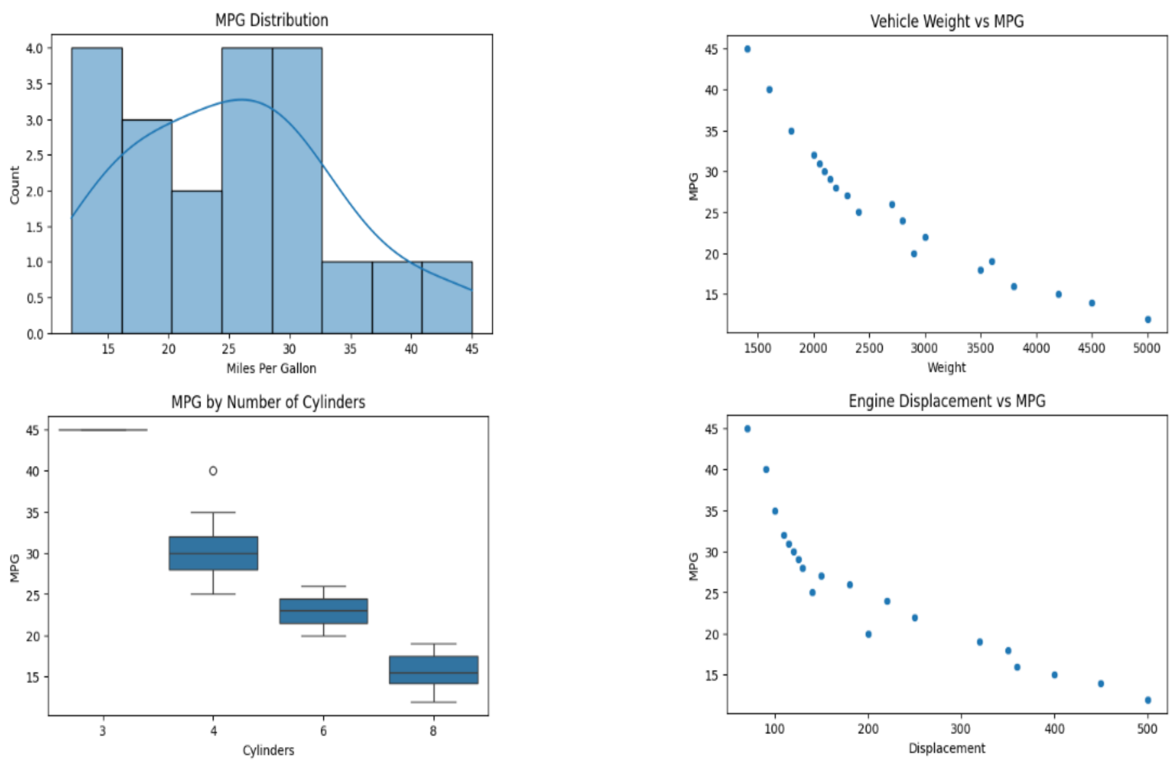
```
sns.histplot(df["MPG"], kde=True)

plt.title("MPG Distribution")

plt.show()

sns.scatterplot(data=df, x="Weight", y="MPG")

plt.title("Weight vs MPG")

plt.show()

sns.heatmap(df.corr(), annot=True)

plt.show()
```

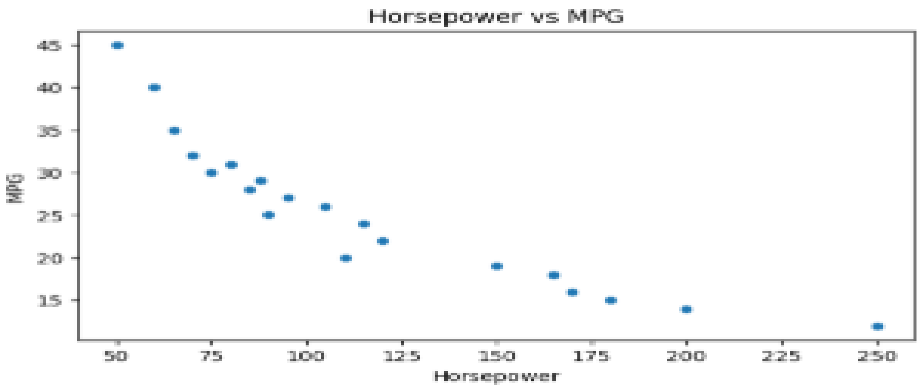
Objective: Investigate vehicle characteristics and their relationship with fuel efficiency.

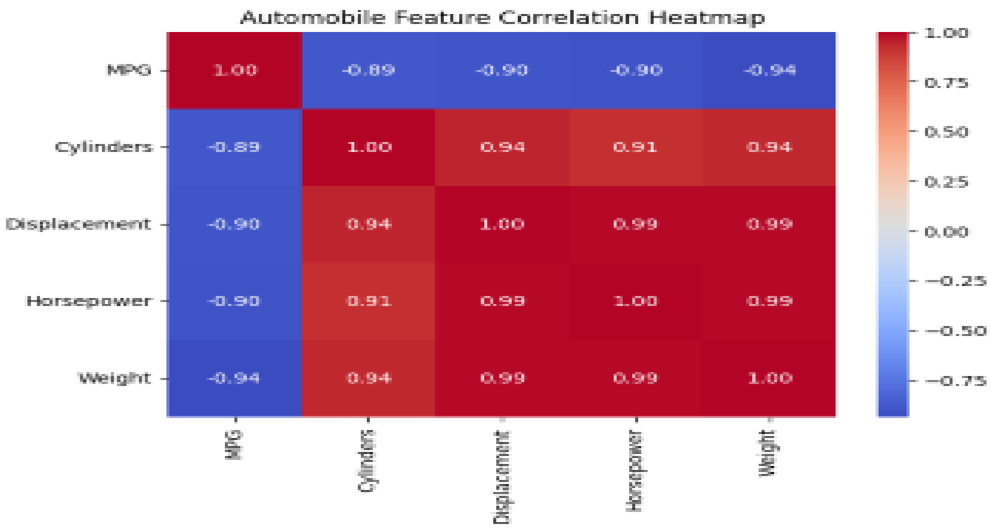
Main analysis:

- Explore the distribution of **MPG (Miles Per Gallon)**.
- Compare mileage across different:
  - Vehicle types
  - Cylinders
  - Model years
  - Origins
- Analyze relationships between:
  - Engine size/displacement vs MPG
  - Vehicle weight vs MPG
  - Horsepower vs MPG
- Detect anomalies and outliers using box plots or IQR.
- Create:
  - Histograms
  - Scatter plots
  - Box plots
  - Correlation heatmaps

Expected finding: Heavier vehicles and vehicles with larger engines generally tend to have lower fuel efficiency, while lighter and smaller-engine vehicles tend to achieve higher MPG.




Horsepower vs MPG
MPG
Horsepower
45
40
35
30
25
20
15
50
75
100
125
150
175
200
225
250


Automobile Feature Correlation Heatmap
MPG
Cylinders
Displacement
Horsepower
Weight
1.00 -0.89 -0.90 -0.90 -0.94
-0.89 1.00 0.94 0.91 0.94
-0.90 0.94 1.00 0.99 0.99
-0.90 0.91 0.99 1.00 0.99
-0.94 0.94 0.99 0.99 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75